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Studierichting: Informatica
Oefeningenreeks 4A nr. 43.10

$$\int \frac{5x^2 + 3x - 2}{2x^3 - 3x^2 + 1} dx$$

Vermits $2x^3 - 3x^2 + 1 = (x - 1)^2(2x + 1)$:

$$\int \frac{5x^2 + 3x - 2}{(x - 1)^2(2x + 1)} dx$$
$$\frac{5x^2 + 3x - 2}{2x^3 - 3x^2 + 1} = \frac{5x^2 + 3x - 2}{(x - 1)^2(2x + 1)} = \frac{A}{(x - 1)^2} + \frac{B}{(x - 1)} + \frac{C}{2x + 1}$$

Voor $x = \frac{1}{2}$:

$$C = \frac{5x^2 + 3x - 2}{(x - 1)^2} = -1$$

Voor $x = 1$:

$$A = \frac{5x^2 + 3x - 2}{2x + 1} = 2$$
$$\frac{5x^2 + 3x - 2}{(x - 1)^2(2x + 1)} - \frac{2}{(x - 1)^2} = \frac{5x^2 + 3x - 2 - 2(2x + 1)}{(x - 1)^2(2x + 1)} = \frac{5x^2 - x - 4}{(x - 1)^2(2x + 1)}$$
$$= \frac{(x - 1)(5x + 4)}{(x - 1)^2(2x + 1)} = \frac{(5x + 4)}{(x - 1)(2x + 1)}$$

Voor $x = 1$:

$$B = \frac{5x + 4}{2x + 1} = 3$$

Splitsing wordt dus:

$$\frac{5x^2 + 3x - 2}{2x^3 - 3x^2 + 1} = \frac{2}{(x - 1)^2} + \frac{3}{(x - 1)} - \frac{1}{2x + 1}$$

Bijgevolg is:

$$I = \int \frac{2}{(x - 1)^2} dx + \int \frac{3}{(x - 1)} dx + \int \frac{1}{2x + 1} dx$$

$$t = x - 1$$

$$dt = dx$$

$$u = 2x + 1$$

$$du = 2dx$$

$$\begin{aligned}
&= 2 \int \frac{1}{t^2} dt + 3 \int \frac{1}{t} dt - \frac{1}{2} \int \frac{1}{u} dt \\
&= -\frac{2}{x-1} + 3 \ln|x-1| - \frac{1}{2} \ln|2x+1| + C
\end{aligned}$$

References